Company Brain: Knowledge That Remains Useful

A thesis of cumulative expertise, bounded agency and human direction

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Abstract

Useful work leaves more than an answer. It leaves evidence, corrections, methods and a better understanding of what matters. This thesis proposes an architecture in which these consequences become durable, scoped resources for later human–agent collaboration. A fast retrieval layer makes current knowledge available; a deliberative layer applies selected procedures and tools; an evaluation boundary determines what may be retained and reused. The central claim is conditional: accumulated experience can increase organisational capability when its applicability, authority and consequences remain inspectable and revisable. We develop that claim into contracts for memory, skills, task execution, scheduled routines and human oversight. We connect the architecture to research on retrieval and experiential agents, then specify an implementation sequence and an evaluation programme. The contribution is a coherent design thesis and a testable method. Its proposed gains in transfer, efficiency and human autonomy remain empirical questions.

1. From an answer to an accumulated capability

A person often repeats context because a previous success has not become usable knowledge. A team repeats a mistake because the correction remained in a conversation. An automated system repeats a procedure because it cannot distinguish an accepted method from an obsolete instruction. These are continuity problems: the consequence of work fails to reach the next relevant decision.

Company Brain treats a completed piece of work as a potential contribution to future capability. The retained unit connects an objective, its scope, the sources consulted, an output, the checks applied and any correction. Later work can recover that unit without reconstructing the whole conversation. A useful method may become a skill when its prerequisites and limits can be stated and its behaviour tested.

The argument has four steps. Work produces experience. Evaluation identifies a useful consequence within a scope. Structured retention makes that consequence available. Appropriate retrieval lets it affect later work. Each step can fail independently. A large archive without selection burdens the next task; a reusable procedure without valid scope spreads error; an apparently successful output without evaluation teaches the wrong lesson.

The intended accumulation belongs to an arrangement of people, models, records and tools. It can occur with fixed model weights. Human learning, model training and improvement of an external procedure are distinct processes, even when they interact during the same project.

2. Research foundations and the proposed contribution

Retrieval-augmented generation connects generated answers to an external information resource [1]. This supports the separation between a model's parameters and task-accessible knowledge. Company Brain extends the design question to authority, correction and the conditions under which an earlier result remains eligible for use.

Reflexion studies the reuse of linguistic feedback across trials [2]. Agent Workflow Memory derives reusable workflows from experience and supplies them during later tasks [3]. These mechanisms support investigating procedural accumulation. Their reported outcomes belong to their respective evaluations; they do not establish the effectiveness of the architecture proposed here.

Provenance standards supply a language for connecting entities, activities and responsible agents [4]. Distributed tracing provides a way to correlate activity across execution boundaries [5]. Together they motivate

inspectable dependencies and receipts. Recording a source identity or a successful call establishes neither the truth of a statement nor the usefulness of the result.

The proposed contribution is the integration of these concerns around continuing work: a versioned relationship between purpose, source, procedure, execution, evaluation and retained consequence. We claim a design synthesis, not priority for memory, retrieval, orchestration or feedback themselves. A stronger research contribution would demonstrate that this integration improves defined outcomes relative to simpler alternatives.

3. Memory as an accountable relationship

Memory should preserve distinctions that matter to action. An episode records what happened. A factual claim records what is supported and by which source. A decision records a choice and its scope. A preference belongs to a person or project. A skill describes a reusable procedure. Operational state records an unfinished or completed activity. These categories may link to one another without being interchangeable.

Every reusable record needs identity, scope, version, provenance and status. A candidate remains a candidate after it is indexed. A superseded rule remains historical evidence while becoming ineligible for current work. A correction should identify affected dependants, including saved answers and skills, rather than merely adding another competing paragraph to the archive.

Retrieval first applies the authority and scope boundary, then relevance. The selected context should contain enough evidence to act and enough uncertainty to abstain. Loading an entire archive can obscure the governing instruction. A compact map instead leads from the relevant area to its current records and supporting material.

4. Two layers with different responsibilities

Layer 1 handles orientation and qualified retrieval: identify the outcome, project, relevant domain, available evidence and eligible saved result. Exact reuse is appropriate only when the question and its significant context still match. Changed sources, instructions or scope require reconsideration. Similar wording alone is insufficient.

Layer 2 performs task-specific deliberation and execution. It receives selected evidence, the exact procedure, actual available tools, permitted effects, output destination and acceptance criteria. It can ask for missing information, perform bounded work or return a useful failure. The layers describe responsibilities; they need not require two model calls for every request.

An accepted Layer 2 result can enrich Layer 1. Retention preserves what was checked, under which conditions, and with what remaining uncertainty. This creates a feedback path without assuming that every fluent response deserves promotion. A saved result can remain useful research while being unsuitable for automatic replay.

5. Skills and organisational structure

A skill combines purpose, prerequisites, steps, resources, tool requirements, output contract and tests. Expertise becomes portable when another executor can determine when the skill applies and how to recognise failure. A text file that describes an unavailable tool is a procedure proposal, not an executable capability.

Three organisational levels clarify responsibility. The executive level maintains purpose, priorities and acceptable trade-offs. The coordinating level resolves dependencies and assigns bounded work. Specialists perform the selected operations and return artifacts with evidence. These functions can be implemented by people, deterministic code or models. Additional agents are justified by useful separation of responsibility, not by the appearance of a corporate hierarchy.

Handoffs identify the source versions, permitted tools, writable location, expected output and acceptance requirements. The receiving step must know whether its prerequisite succeeded. A final summary cannot repair a missing intermediate result. Independent checking is valuable where it can detect different errors; repeated agreement from the same evidence does not create independence.

6. MAPS and the visible system

MAPS offers four useful questions. Memory: where is the relevant knowledge? Agent: who or what can carry out the work? Pulse: what initiates recurring work? Screen: what can a person understand and control? These are design views over shared records, not separate truths.

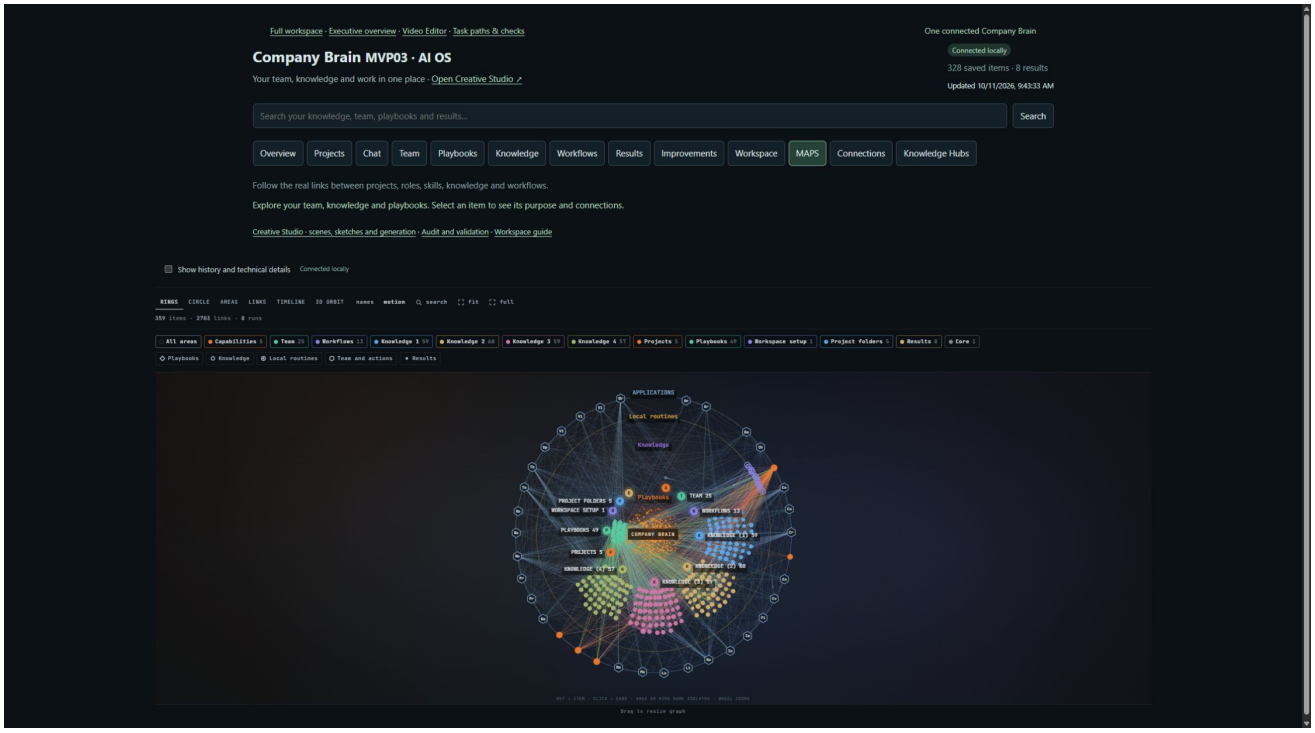


Figure 1. An observed Company Brain MAPS overview, captured with item names and technical details hidden. The image illustrates navigation among areas and record types. A displayed connection means a recorded relationship; it does not certify that a workflow ran. Knowledge groups are display partitions, not separate minds. The conceptual figures in the annexes explain selected paths without exposing private records.

A useful screen answers: what happened, where is the result, why does it matter, and what requires a decision? Names should describe functions. Links should lead to the current record or its preserved predecessor. A graph supports inspection; it becomes useful through the quality and meaning of its relationships rather than the number of edges.

7. Routines, pulses and bounded improvement

A routine is a defined procedure. A pulse supplies a trigger. An execution is an owned attempt to carry out a task. These distinctions prevent a configured schedule from being mistaken for completed work. A manual button and a scheduled occurrence should reach the same execution contract.

The execution loop observes current state, chooses an allowed action, performs one step, retains a receipt and checks the result. It then continues, repairs, waits, reconciles an uncertain effect or finishes. Attempts, time and authority are bounded. A timeout after an external effect requires reconciliation using the original operation identity; repeating the effect can create a duplicate.

Improvement follows a separate loop: collect useful outcomes and failures, propose a specific procedural change, test it against development cases and previously retained failures, then evaluate transfer on held-out

tasks. Promotion requires evidence suitable for the claimed improvement. A rewritten instruction is a candidate change until its consequences have been checked.

Passive operation means that routine authorised work proceeds without unnecessary intervention and returns meaningful outcomes. Human attention is reserved for exceptions and decisions that exceed delegated authority. Availability, recovery, valid tool access and notification behaviour must be tested. A timer alone establishes none of them.

8. Human direction and consent

The companion thesis, The Architecture of Consent, asks who shapes the conditions under which choices are made. In Company Brain, that question applies to source selection, skill promotion and the objectives used by evaluators. An orderly system can faithfully preserve an unsound judgment.

The proposed response is contestability: people can inspect an accepted rule, change its scope, challenge a source, revise an objective and see which future operations are affected. A record of approval is useful only when the person understands what was approved. Retained rationale should be a concise, accountable explanation of the decision and evidence, not an exposure of private deliberation.

Technical correctness and legitimate authority require distinct checks. A task can produce the requested artifact and still serve an objective that its owner wishes to revise. The design therefore preserves both execution evidence and the route by which governing direction can change.

9. Implementation derived from the argument

Begin with one domain, one owner and one reversible workflow. Store versioned records and make their current eligibility queryable. Add a bounded retrieval path, then a task contract and a deterministic runner. Introduce model decisions only where the next step genuinely depends on interpretation. Add artifact checks before adding recurring triggers.

A practical local stack can use ordinary files for documents, a transactional database for current state and an append-only history for changes. Choose one authoritative writer and rebuildable views. Product artifacts remain in their own workspaces; the control layer retains links and bounded summaries. Access enforcement belongs in the runtime and tools, not solely in written instructions.

Introduce a public synthetic example before connecting valuable data. Demonstrate correction propagation, rejected stale writes, missing evidence and duplicate triggers. Add external adapters individually, verifying both their interfaces and actual results. Move to a distributed execution service when the requirements include durable waits, concurrent ownership or remote recovery that the local design cannot reliably provide.

The adoption guide and annexes specify the sequence in more detail. They are portable design instructions rather than an installation manual for a proprietary product.

10. Evaluation and scholarly development

The primary empirical question is whether selected, validated procedures improve accepted work beyond access to the same factual knowledge. Compare three conditions: task context alone; task context plus accepted facts; and the same facts plus validated procedures. Hold model, tools, permissions and budgets constant. Freeze the memory for each offline trial and isolate task-family splits.

Measure accepted outputs, unsupported claims, stale-rule use, human correction time, latency and actual cost across all attempts. Report unsuccessful tasks as well as successful ones. Use repeated trials for stochastic execution and uncertainty estimates appropriate to related task families. Separate evaluation of online correction from frozen-memory comparisons.

The public demonstration accompanying this edition is a deterministic teaching fixture. Historical engineering work informs the design; neither fixture success nor regression counts establish a model-performance effect.

The research programme requires a published task set, meaningful baselines, ablations of individual safeguards, independent outcome assessment where feasible, and reporting of contrary results.

Conclusion

Company Brain proposes that organisational capability can accumulate through recoverable, evaluated consequences of work. Its organising unit is the relationship between purpose, evidence, procedure and accepted result. Memory makes that relationship available; bounded execution applies it; correction keeps it revisable; human direction determines what it is for. The thesis becomes useful through implementations and experiments that expose where those relationships hold and where they fail.

References

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Preparation note

Generative AI assisted source discovery, editing, document preparation and the synthetic demonstration. The named author remains responsible for the thesis. This edition makes no claim of peer review or independent experimental validation.